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Practice Assignment • 15 min

Due Jun 29, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. Let $S = \{1, 2, 3\}$. Which one of the following sets represents $\mathcal{P}(S)$? 1 / 1 point

- {{1}, {2}, {3}, {1,2}, {1,3}, {2,3}, {1,2,3}}
- $\{\{1\}, \{2\}, \{3\}\}$
- {{1}, {2}, {3}, {1,2}, {1,3}, {2,3}}
- {$\emptyset$, {1}, {2}, {3}, {1,2}, {1,3}, {2,3}, {1,2,3}}

Nice work

This set contains all the subsets of S

2. Given a set A of 10 elements. What is the number of elements in the powerset of A? 1 / 1 point

1024

Nice work

$|\mathcal{P}(A)| = 2^{|A|} = 2^{10} = 1024$

3. Given a set A of 8 elements, what is the cardinality of the powerset of A, $|\mathcal{P}(A)|$? 1 / 1 point

256

Nice work

$|\mathcal{P}(A)| = 2^{|A|} = 2^8 = 256$

4. Let S be a set. If $|S| = 11$ then $|\mathcal{P}(S)|$ is: 1 / 1 point

2048

Nice work

$|\mathcal{P}(S)| = 2^{|S|} = 2^{11} = 2048$

5. Let S be a set. If $|\mathcal{P}(S)| = 16$ then $|S|$ is: 1 / 1 point

4

Nice work

$|\mathcal{P}(S)| = 16 = 2^4 = 2^{|S|}$ implies $|S| = 4$

6. Let $S = \{\{a, b\}, \{c\}\}$. The number of elements in $\mathcal{P}(S)$ is: 1 / 1 point

- 8
- 4
- 3
- 6

Nice work

$|S| = 2$, hence, the number of elements in $\mathcal{P}(S)$ is equal to $2^2 = 4$